

SYSTEM DESIGN DOCUMENT

Project Title

SwapSphere – A Smart Barter Platform with Intelligent Exchange Matching

1. Introduction

1.1 Purpose

This document describes the overall system design of SwapSphere. It explains the architecture, major system modules, component interactions, and workflow followed by the application. The purpose of this document is to provide a high-level understanding of how the system is organized before implementation.

1.2 System Overview

SwapSphere is a web-based barter platform developed using the MERN stack. The application enables users to exchange items without monetary transactions through both direct exchanges and intelligent graph-based exchange matching.

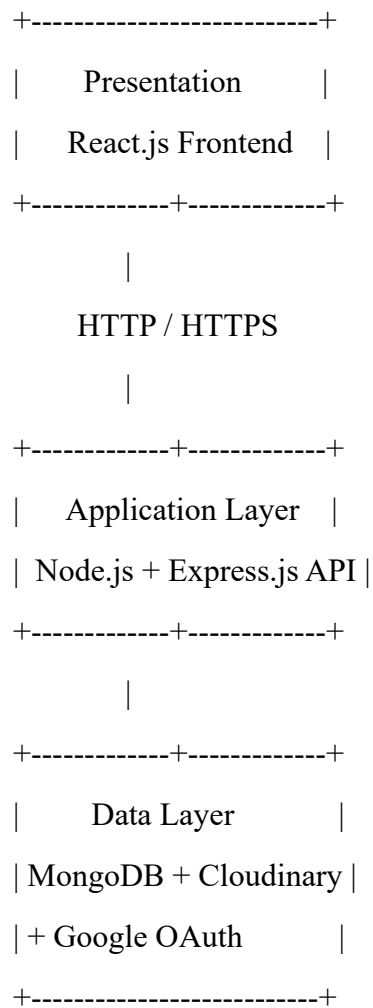
The system consists of three major layers:

- Presentation Layer
- Application Layer
- Data Layer

Each layer performs specific responsibilities while communicating with the others through well-defined interfaces.

2. System Architecture

SwapSphere follows a **Three-Tier Client–Server Architecture**.



2.1 Presentation Layer

The Presentation Layer provides the graphical user interface through which users interact with the system.

Responsibilities

- User Registration
- Login
- Marketplace Browsing
- User Profile
- Item Listing
- Exchange Suggestions
- Exchange Requests

- Exchange Room
- Notifications
- Administrator Dashboard

Technologies Used

- React.js
- Bootstrap

2.2 Application Layer

The Application Layer contains the complete business logic of SwapSphere.

Responsibilities

- Authentication
- Authorization
- Item Management
- Graph Matching
- Exchange Request Management
- Exchange Room Management
- Messaging
- Notification Management
- Rating Calculation
- Administrative Operations

Technologies Used

- Node.js
- Express.js

2.3 Data Layer

The Data Layer is responsible for storing and retrieving application data.

Responsibilities

- Store user information
- Store item listings
- Store exchange requests

- Store Exchange Rooms
- Store messages
- Store notifications
- Store reviews
- Store categories and subcategories

External Services

- MongoDB
- Cloudinary
- Google OAuth

3. System Modules

3.1 Authentication Module

The Authentication Module manages user registration, login, logout, and session management. Users can authenticate using email/password or Google OAuth.

3.2 User Management Module

This module manages user profiles, profile updates, profile pictures, trust indicators, exchange history, and reviews.

3.3 Marketplace Module

The Marketplace Module displays all publicly available item listings.

Users can

- Search listings
- Filter by category
- Filter by subcategory
- View item details
- Initiate direct exchanges

3.4 Item Management Module

The Item Management Module allows registered users to

- Create listings

- Edit listings
- Delete listings
- Upload images
- Upload videos
- Select category
- Select optional subcategory
- Specify prioritized exchange preferences

3.5 Graph Matching Module

The Graph Matching Module analyzes all AVAILABLE listings and identifies possible exchange opportunities.

The module:

- Detects two-way exchanges.
- Detects three-way exchange cycles.
- Ranks direct exchanges separately.
- Ranks three-way exchanges separately.
- Calculates a combined exchange score for three-way exchanges.
- Displays only suggestions. Users decide whether to initiate an exchange.

3.6 Exchange Request Module

This module manages exchange requests created from

- Graph-based suggestions
- Direct marketplace browsing

It handles

- Request creation
- Acceptance
- Rejection
- Cancellation
- Expiration

3.7 Exchange Room Module

After every participant accepts an exchange request, an Exchange Room is automatically created.

The Exchange Room allows participants to

- Chat
- Share images
- Schedule meetings
- Confirm exchange completion

3.8 Messaging Module

Provides communication between exchange participants.

Supported message types

- Text
- Image
- System Messages

3.9 Notification Module

Generates notifications for

- Exchange requests
- Exchange suggestions
- Accepted requests
- Rejected requests
- Cancelled requests
- Expired requests
- Meeting updates
- Exchange completion

3.10 Rating and Review Module

Allows users to

- Rate exchange partners
- Write reviews

- Update trust indicators

3.11 Administration Module

Allows administrators to

- Manage users
- Manage categories
- Manage listings
- Monitor exchange activities

4. System Workflow

The overall workflow of SwapSphere is illustrated below.

User Registration/Login



Create Item Listing

(Category + Optional Subcategory)



Specify Exchange Preferences

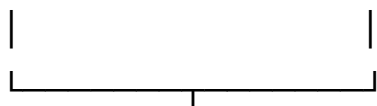
(Priority Order)

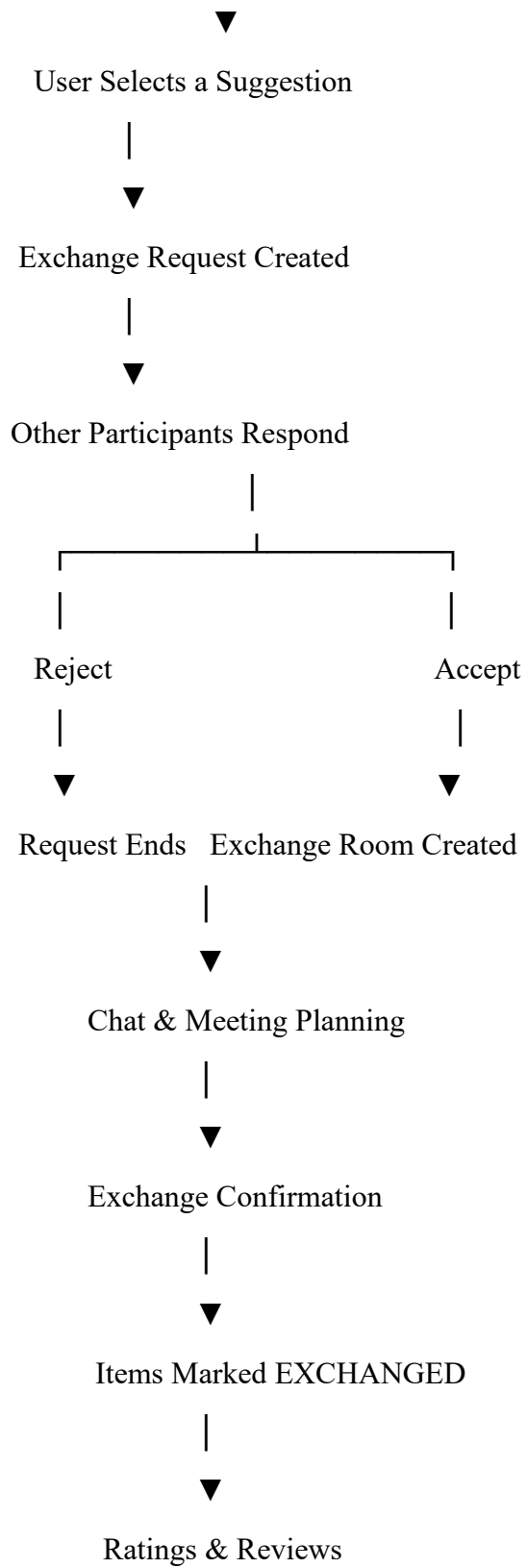


Graph Matching Engine



Direct Suggestions Three-Way Suggestions





5. Design Principles

The system is designed using the following principles:

Modular Design

Each module performs a specific responsibility, making the system easier to maintain and extend.

Separation of Concerns

The frontend, backend, and database are separated to improve maintainability.

Scalability

The architecture allows future support for additional exchange types and larger exchange cycles.

Maintainability

Each module can be modified independently without affecting unrelated modules.

Security

Authentication and authorization ensure that only authorized users can access protected resources.

6. Advantages of the Proposed Design

The proposed design provides several benefits:

- Modular and organized architecture.
- Easy maintenance and future enhancements.
- Efficient graph-based exchange matching.
- Secure authentication and authorization.
- Clear separation between direct exchanges and graph-based suggestions.
- Support for future expansion to N-way exchanges.
- Improved user trust through ratings, reviews, and exchange history.

7. Conclusion

The proposed system design provides a scalable and modular architecture for the development of SwapSphere. The separation of the application into independent modules simplifies development, testing, and maintenance while supporting future enhancements. The graph-based exchange matching mechanism, together with traditional marketplace browsing, provides users with multiple ways to discover suitable barter opportunities, making the platform both flexible and efficient.